

Token Cross-Modal Enhancer: Multi-Scale Spatial-Spectral Transformer for Hyperspectral Image Segmentation

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Abstract

This document presents the Token Cross-Modal Enhancer (TCME), a novel transformer architecture designed for hyperspectral image segmentation. The system processes spatial and spectral modalities through differentiated tokenization strategies, multi-scale hierarchical attention, and intelligent token compression mechanisms to achieve efficient and accurate pixel-wise classification for medical hyperspectral imaging applications. Building upon modern vision transformer architectures and cross-modal attention mechanisms, TCME addresses the fundamental challenges of token imbalance and computational complexity in dual-modal processing while maintaining high segmentation accuracy.

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1 Introduction

Hyperspectral imaging (HSI) captures both spatial geometry and spectral signatures across multiple wavelength bands, providing rich information for material identification and classification [6]. In medical applications, this dual-modal information enables precise tissue characterization and pathological boundary detection. However, the high dimensionality and computational complexity of HSI data pose significant challenges for deep learning architectures [7].

The Token Cross-Modal Enhancer addresses these challenges through:

- **Modality-specific tokenization:** Optimized representation for spatial and spectral characteristics inspired by vision transformer tokenization strategies [1]
- **Multi-scale hierarchical processing:** Capturing fine details and global context simultaneously using hierarchical token division [2]
- **Cross-modal attention mechanisms:** Learning spatial-spectral relationships across different granularities through bidirectional attention [3]
- **Intelligent token compression:** Maintaining informative tokens while reducing computational overhead via advanced pruning techniques [4]

2 Pipeline Architecture Overview

Before diving into detailed components, we present the complete Token Cross-Modal Enhancer architecture:

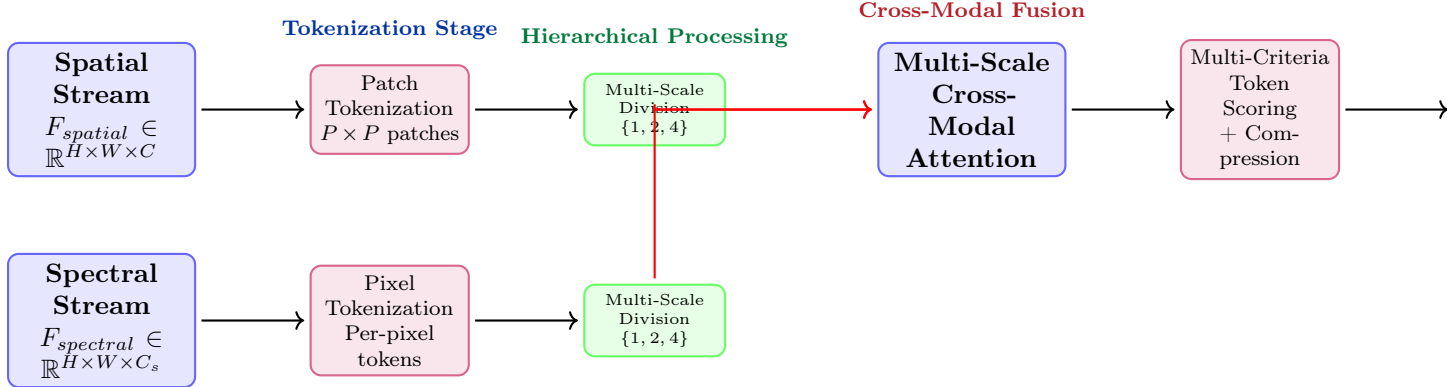


Figure 1: Token Cross-Modal Enhancer pipeline overview showing the complete architecture from dual-modal input processing to enhanced token generation.

Intuition: Why Token-Based Cross-Modal Processing?

Token-based processing offers several advantages for hyperspectral data: (1) **Modality-specific representation** allows optimized encoding for spatial geometry versus spectral signatures, (2) **Scalable attention** enables efficient processing of high-dimensional data through token reduction, and (3) **Cross-modal alignment** preserves spatial correspondence between different tokenization granularities through shared positional encoding [5].

3 Input Processing and Tokenization Strategy

3.1 Spatial Stream Input Processing

Input: Fused spatial features $F_{\text{spatial}} \in \mathbb{R}^{H \times W \times C}$ from upstream processing pipeline including:

- Local spatial features via 3D→2D CNN cascade for spectral-spatial feature extraction
- Global spatial context via 2D Mamba blocks for long-range dependency modeling
- Dynamic weighting and leaky-gated fusion for adaptive feature combination

Tokenization Strategy: Spatial Patch Tokenization Following the successful paradigm established by Vision Transformers [1], we employ patch-based tokenization for spatial features.

Mathematical Formulation: Given input $F_{\text{spatial}} \in \mathbb{R}^{H \times W \times C}$, patch tokenization proceeds as:

$$\text{Patches} = \text{Unfold}(F_{\text{spatial}}, P) \in \mathbb{R}^{N_{\text{patches}} \times P^2 C} \quad (1)$$

$$N_{\text{patches}} = \frac{H}{P} \times \frac{W}{P} \quad (2)$$

$$T_{\text{spatial}}^{(0)} = \text{Patches} \cdot W_{\text{tok}}^{\text{spatial}} + b_{\text{spatial}} \quad (3)$$

$$T_{\text{spatial}}^{(1)} = T_{\text{spatial}}^{(0)} + \text{PE}_{2D}(\text{patch_coords}) \quad (4)$$

where $W_{\text{tok}}^{\text{spatial}} \in \mathbb{R}^{P^2 C \times D}$ is the learned linear projection and PE_{2D} is 2D sinusoidal positional encoding.

Rationale for Patch Tokenization [1, 2]:

1. **Computational Efficiency:** Reduces token count from $H \times W$ to $(H/P) \times (W/P)$, enabling quadratic complexity reduction in attention mechanisms
2. **Spatial Context Preservation:** Each patch token captures local spatial patterns while maintaining neighborhood relationships
3. **Established Effectiveness:** Proven performance in vision transformers for segmentation tasks, particularly in hierarchical architectures

3.2 Spectral Stream Input Processing

Input: Spectral feature map $F_{\text{spectral}} \in \mathbb{R}^{H \times W \times C_s}$ where each (i, j) location contains compressed spectral signatures from approximately 30 bands (reduced from original 38 bands through spectral preprocessing and dimensionality reduction).

Tokenization Strategy: Spectral Pixel Tokenization For spectral data, we employ pixel-level tokenization to preserve fine-grained spectral information crucial for material discrimination.

Mathematical Formulation:

$$\text{Pixels} = \text{Reshape}(F_{\text{spectral}}, H \times W, C_s) \quad (5)$$

$$T_{\text{spectral}}^{(0)} = \text{Pixels} \cdot W_{\text{tok}}^{\text{spectral}} + b_{\text{spectral}} \quad (6)$$

$$T_{\text{spectral}}^{(1)} = T_{\text{spectral}}^{(0)} + \text{PE}_{2D}(\text{pixel_coords}) \quad (7)$$

where $W_{\text{tok}}^{\text{spectral}} \in \mathbb{R}^{C_s \times D}$ projects pixel-level spectral features to common embedding dimension D .

Rationale for Pixel Tokenization [6]:

1. **Fine-Grained Spectral Information:** Preserves pixel-level spectral signatures crucial for material discrimination and tissue boundary detection
2. **Medical HSI Requirements:** Tissue boundaries often exhibit sharp spectral transitions requiring pixel-level precision for accurate segmentation
3. **Spectral Signature Preservation:** Maintains complete spectral information without spatial averaging that could blur important material distinctions

3.3 Synchronization Through Shared Positional Encoding

Key Innovation: Both modalities use the same 2D sinusoidal positional encoding function but at different granularities, enabling natural cross-modal alignment despite different tokenization strategies.

Mathematical Formulation: The 2D sinusoidal positional encoding is defined as:

$$\text{PE}_{2D}(x, y)[2d] = \sin\left(\frac{x}{10000^{2d/D}}\right) \quad (8)$$

$$\text{PE}_{2D}(x, y)[2d + 1] = \cos\left(\frac{x}{10000^{2d/D}}\right) \quad (9)$$

$$\text{PE}_{2D}(x, y)[2d + D] = \sin\left(\frac{y}{10000^{2d/D}}\right) \quad (10)$$

$$\text{PE}_{2D}(x, y)[2d + 1 + D] = \cos\left(\frac{y}{10000^{2d/D}}\right) \quad (11)$$

where:

- **Spatial tokens:** PE encodes patch-center coordinates (x_p, y_p) where $(x_p, y_p) = (i \cdot P + P/2, j \cdot P + P/2)$
- **Spectral tokens:** PE encodes exact pixel coordinates $(x_i, y_i) = (i, j)$
- **Cross-attention alignment:** Spatial patches naturally attend to spectral pixels within their spatial region due to position similarity in embedding space

Intuition: Shared Positional Encoding Benefits

This creates a continuous positional space where tokens at nearby spatial locations have similar positional embeddings, enabling natural cross-modal alignment. The shared PE allows the model to understand spatial relationships between different tokenization granularities without additional alignment mechanisms [5].

4 Multi-Scale Token Division

4.1 Hierarchical Scale Generation

Following hierarchical vision transformer design principles [2, 3], both spatial and spectral token sets are divided into three scales $\{1, 2, 4\}$ to capture multi-resolution patterns and enable efficient attention computation.

Spatial Multi-Scale Division:

$$T_{\text{spatial}}^{(1)} : [N_{\text{patches}}, D] \quad \# \text{Full resolution} \quad (12)$$

$$T_{\text{spatial}}^{(2)} : \left[\frac{N_{\text{patches}}}{4}, D \right] \quad \# 2 \times 2 \text{ average pooling} \quad (13)$$

$$T_{\text{spatial}}^{(4)} : \left[\frac{N_{\text{patches}}}{16}, D \right] \quad \# 4 \times 4 \text{ average pooling} \quad (14)$$

Spectral Multi-Scale Division:

$$T_{\text{spectral}}^{(1)} : [H \times W, D] \quad \# \text{Full resolution} \quad (15)$$

$$T_{\text{spectral}}^{(2)} : \left[\frac{H \times W}{4}, D \right] \quad \# 2 \times 2 \text{ spatial pooling} \quad (16)$$

$$T_{\text{spectral}}^{(4)} : \left[\frac{H \times W}{16}, D \right] \quad \# 4 \times 4 \text{ spatial pooling} \quad (17)$$

Pooling Operations: For spatial tokens, average pooling over spatial neighborhoods:

$$T_{\text{spatial}}^{(s)}[i] = \frac{1}{s^2} \sum_{k=0}^{s-1} \sum_{l=0}^{s-1} T_{\text{spatial}}^{(1)}[s \cdot i + k \cdot \sqrt{N_{\text{patches}}} + l]$$

For spectral tokens, spatial average pooling maintaining spectral information:

$$T_{\text{spectral}}^{(s)}[i, j] = \frac{1}{s^2} \sum_{m=0}^{s-1} \sum_{n=0}^{s-1} T_{\text{spectral}}^{(1)}[s \cdot i + m, s \cdot j + n]$$

Benefits of Multi-Scale Processing [8]:

- **Fine Scale (1):** Captures detailed patterns, boundaries, and fine-grained features
- **Medium Scale (2):** Balances local detail with contextual information
- **Coarse Scale (4):** Provides global semantic understanding and computational efficiency

4.2 Cross-Scale Attention Pairs

All possible scale combinations $(a, b) \in \{1, 2, 4\}^2$ are processed, creating 9 attention combinations for comprehensive multi-scale interaction:

- **Intra-scale:** (1,1), (2,2), (4,4) — Same resolution interactions for scale-consistent processing
- **Cross-scale:** (1,2), (1,4), (2,1), (2,4), (4,1), (4,2) — Multi-resolution interactions for hierarchical feature fusion

This comprehensive interaction enables fine-to-coarse and coarse-to-fine information flow, similar to Feature Pyramid Networks but within the transformer attention mechanism [9].

5 Attention Mechanisms

5.1 Self-Attention Processing

For each scale pair (a, b) , both modalities undergo self-attention to capture intra-modal dependencies at different scales.

Spatial Self-Attention:

$$Q_{\text{spatial}}^{(a)} = T_{\text{spatial}}^{(a)} W_Q^{\text{spatial}} \quad (18)$$

$$K_{\text{spatial}}^{(a)} = T_{\text{spatial}}^{(a)} W_K^{\text{spatial}} \quad (19)$$

$$V_{\text{spatial}}^{(a)} = T_{\text{spatial}}^{(a)} W_V^{\text{spatial}} \quad (20)$$

$$\text{SA}_{\text{spatial}}^{(a)} = \text{softmax} \left(\frac{Q_{\text{spatial}}^{(a)} K_{\text{spatial}}^{(a)T}}{\sqrt{d_k}} \right) V_{\text{spatial}}^{(a)} \quad (21)$$

Spectral Self-Attention:

$$Q_{\text{spectral}}^{(b)} = T_{\text{spectral}}^{(b)} W_Q^{\text{spectral}} \quad (22)$$

$$K_{\text{spectral}}^{(b)} = T_{\text{spectral}}^{(b)} W_K^{\text{spectral}} \quad (23)$$

$$V_{\text{spectral}}^{(b)} = T_{\text{spectral}}^{(b)} W_V^{\text{spectral}} \quad (24)$$

$$\text{SA}_{\text{spectral}}^{(b)} = \text{softmax} \left(\frac{Q_{\text{spectral}}^{(b)} K_{\text{spectral}}^{(b)T}}{\sqrt{d_k}} \right) V_{\text{spectral}}^{(b)} \quad (25)$$

where d_k is the key dimension and separate projection matrices are used for each modality to capture modality-specific patterns.

5.2 Cross-Modal Attention Mechanisms

Bidirectional Cross-Attention: Cross-modal attention enables each modality to gather relevant information from the other, creating rich inter-modal feature representations.

Spatial $\rightarrow$ Spectral Attention:

$$Q_s^{(a)} = T_{\text{spatial}}^{(a)} W_Q^{\text{cross}} \quad (26)$$

$$K_{sp}^{(b)} = T_{\text{spectral}}^{(b)} W_K^{\text{cross}} \quad (27)$$

$$V_{sp}^{(b)} = T_{\text{spectral}}^{(b)} W_V^{\text{cross}} \quad (28)$$

$$\text{CA}_{s \rightarrow sp}^{(a,b)} = \text{softmax} \left(\frac{Q_s^{(a)} K_{sp}^{(b)T}}{\sqrt{d_k}} \right) V_{sp}^{(b)} \quad (29)$$

Spectral $\rightarrow$ Spatial Attention:

$$Q_{sp}^{(b)} = T_{\text{spectral}}^{(b)} W_Q^{\text{cross}} \quad (30)$$

$$K_s^{(a)} = T_{\text{spatial}}^{(a)} W_K^{\text{cross}} \quad (31)$$

$$V_s^{(a)} = T_{\text{spatial}}^{(a)} W_V^{\text{cross}} \quad (32)$$

$$\text{CA}_{sp \rightarrow s}^{(b,a)} = \text{softmax} \left(\frac{Q_{sp}^{(b)} K_s^{(a)T}}{\sqrt{d_k}} \right) V_s^{(a)} \quad (33)$$

Key Properties:

- **Asymmetric Token Counts:** Attention matrices have different shapes $[N_{\text{spatial}}^{(a)} \times N_{\text{spectral}}^{(b)}]$ and $[N_{\text{spectral}}^{(b)} \times N_{\text{spatial}}^{(a)}]$, handled efficiently through batched matrix operations
- **Position-Guided Attention:** Shared 2D PE enables spatial proximity bias in cross-modal attention computation
- **Modality-Specific Parameters:** Separate projection matrices for spatial and spectral modalities allow specialized feature extraction

Intuition: Cross-Modal Attention Benefits

Cross-modal attention allows spatial tokens to gather spectral information from spatially relevant regions and vice versa. This bidirectional flow enables the model to create spatially-informed spectral representations and spectrally-informed spatial representations, crucial for accurate tissue segmentation [3].

6 Multi-Criteria Token Scoring

6.1 Individual Criterion Computation

For each token i , we compute importance scores from multiple complementary sources to ensure robust token selection.

Attention-Based Scores:

$$S_{\text{sp}}(i) = \sum_{h=1}^H \sum_j \alpha_{\text{sp}}^{(h)}(i, j) \quad \# \text{Spatial self-attention aggregation} \quad (34)$$

$$S_{\text{spc}}(i) = \sum_{h=1}^H \sum_j \alpha_{\text{spc}}^{(h)}(i, j) \quad \# \text{Spectral self-attention aggregation} \quad (35)$$

$$S_{s \rightarrow sp}(i) = \sum_{h=1}^H \sum_j \alpha_{s \rightarrow sp}^{(h)}(i, j) \quad \# \text{Spatial} \rightarrow \text{Spectral cross-attention} \quad (36)$$

$$S_{sp \rightarrow s}(i) = \sum_{h=1}^H \sum_j \alpha_{sp \rightarrow s}^{(h)}(i, j) \quad \# \text{Spectral} \rightarrow \text{Spatial cross-attention} \quad (37)$$

where H is the number of attention heads and $\alpha^{(h)}(i, j)$ represents attention weights from the h -th head.

Value Vector Magnitude:

$$V_{\text{norm}}(i) = \|v_i\|_1 = \sum_{d=1}^D |v_i[d]|$$

Rationale for Value Norm Inclusion [10, 11]: Recent research demonstrates that attention scores alone are insufficient for token importance evaluation. Value vector norms capture the actual informational content being transmitted, preventing retention of "attention sink" tokens that receive high attention but contribute minimally to the output representation.

Gradient-Based Importance (Optional Enhancement):

$$S_{\text{grad}}(i) = \|\nabla_{v_i} \mathcal{L}\|_2$$

where $\mathcal{L}$ is the task-specific loss function, providing gradient-based importance estimation.

6.2 Combined Scoring Function

Multiplicative Fusion with Learnable Temperatures:

$$S_{\text{tok}}(i) = \prod_{k=1}^5 [S_k(i)]^{\tau_k}$$

where $S_k(i) \in \{S_{\text{sp}}(i), S_{\text{spc}}(i), S_{s \rightarrow \text{sp}}(i), S_{\text{sp} \rightarrow s}(i), V_{\text{norm}}(i)\}$ and τ_k are learnable temperature parameters.

Temperature Learning: The temperature parameters are learned through gradient descent:

$$\tau_k^{(t+1)} = \tau_k^{(t)} - \eta \frac{\partial \mathcal{L}_{\text{task}}}{\partial \tau_k}$$

where $\mathcal{L}_{\text{task}}$ is the downstream segmentation loss, allowing adaptive weighting of different scoring criteria.

7 Token Compression Strategies

7.1 Joint Pair-Constrained Selection (Primary Strategy)

To preserve cross-modal alignment while reducing computational overhead, we employ a two-stage compression approach.

Stage 1: Cross-Modal Pair Selection We first select spatial-spectral token pairs with the highest mutual attention scores:

$$S_{\text{pair}}(i, j) = A_{s \rightarrow \text{sp}}(i, j) + A_{\text{sp} \rightarrow s}(j, i) + \text{PE_sim}(i, j)$$

where $\text{PE_sim}(i, j) = \exp(-\|\text{PE}_i - \text{PE}_j\|_2^2 / 2\sigma^2)$ adds spatial proximity bias.

The optimization problem is:

$$\mathcal{P}^* = \arg \max_{\mathcal{P} \subseteq \{(i, j)\}} \sum_{(i, j) \in \mathcal{P}} S_{\text{pair}}(i, j) \quad \text{s.t.} \quad |\mathcal{P}| \leq N_{\text{pairs}}$$

This is solved greedily by iteratively selecting the pair with highest score while maintaining disjoint sets.

Algorithm 1 Joint Pair-Constrained Selection

```

1: Input:  $S_{\text{pair}}, N_{\text{pairs}}$ 
2: Initialize  $\mathcal{P} = \emptyset, \mathcal{S} = \emptyset, \mathcal{T} = \emptyset$ 
3: while  $|\mathcal{P}| < N_{\text{pairs}}$  do
4:    $(i^*, j^*) = \arg \max_{(i,j) \notin (\mathcal{S} \times \mathcal{T})} S_{\text{pair}}(i, j)$ 
5:    $\mathcal{P} = \mathcal{P} \cup \{(i^*, j^*)\}$ 
6:    $\mathcal{S} = \mathcal{S} \cup \{i^*\}, \mathcal{T} = \mathcal{T} \cup \{j^*\}$ 
7: end while
8: Return:  $\mathcal{P}, \mathcal{S}, \mathcal{T}$ 

```

Algorithm: Greedy Pair Selection**7.2 Top-K Selection Within Paired Tokens (Secondary Refinement)**

Within the paired subsets $\mathcal{S}$ and $\mathcal{T}$, we perform final token selection based on individual importance scores:

$$\begin{aligned}\tilde{\mathcal{S}} &= \text{TopK}(\{S_{\text{tok}}(i) : i \in \mathcal{S}\}, K_{\text{spatial}}) \\ \tilde{\mathcal{T}} &= \text{TopK}(\{S_{\text{tok}}(j) : j \in \mathcal{T}\}, K_{\text{spectral}})\end{aligned}$$

Adaptive Token Budget Strategy:

$$N_{\text{pairs}} = 5000 \quad \# \text{Initial pair constraint} \quad (38)$$

$$K_{\text{spatial}} = 800 \quad \# \text{Final spatial token count} \quad (39)$$

$$K_{\text{spectral}} = 2000 \quad \# \text{Final spectral token count} \quad (40)$$

This yields a final token ratio of approximately 2.5 : 1 (spectral:spatial), significantly more balanced than the original 256 : 1 imbalance.

7.3 Mathematical Justification

The two-stage optimization can be formulated as:

Stage 1 Objective:

$$\max_{\mathcal{P}: |\mathcal{P}| \leq N_{\text{pairs}}} \sum_{(i,j) \in \mathcal{P}} S_{\text{pair}}(i, j)$$

Stage 2 Objectives:

$$\begin{aligned}\max_{\mathcal{S}' \subseteq \mathcal{S}, |\mathcal{S}'| = K_{\text{spatial}}} \sum_{i \in \mathcal{S}'} S_{\text{tok}}(i) \\ \max_{\mathcal{T}' \subseteq \mathcal{T}, |\mathcal{T}'| = K_{\text{spectral}}} \sum_{j \in \mathcal{T}'} S_{\text{tok}}(j)\end{aligned}$$

This formulation guarantees that both inter-modal alignment (Stage 1) and intra-modal quality (Stage 2) are optimized simultaneously.

Intuition: Two-Stage Compression Benefits

The two-stage approach ensures: (1) **Alignment preservation** by prioritizing cross-modal relationships, (2) **Computational efficiency** through aggressive yet informed token reduction, (3) **Quality assurance** by combining relationship strength with individual importance, and (4) **Balanced representation** preventing orphaned tokens without cross-modal partners [4].

8 Output Generation and Integration

The Token Cross-Modal Enhancer outputs two compressed token sets with preserved positional information:

$$\tilde{T}_{\text{spatial}} \in \mathbb{R}^{K_{\text{spatial}} \times D}, \quad \tilde{T}_{\text{spectral}} \in \mathbb{R}^{K_{\text{spectral}} \times D}$$

Each token retains its 2D positional encoding PE_{2D} , enabling spatial reconstruction and downstream integration:

$$\begin{aligned} \tilde{T}_{\text{spatial}}[i] &= [\text{feature}_i; \text{PE}_{2D}(\text{pos}_i)] \\ \tilde{T}_{\text{spectral}}[j] &= [\text{feature}_j; \text{PE}_{2D}(\text{pos}_j)] \end{aligned}$$

Downstream Integration Options:

- **Concatenation:** $T_{\text{fused}} = [\tilde{T}_{\text{spatial}}; \tilde{T}_{\text{spectral}}]$
- **Cross-attention:** Additional cross-attention layers for further fusion
- **Decoder Integration:** Direct integration into transformer decoder architectures

9 Addressing Key Challenges

9.1 Token Count Imbalance

Challenge: Pixel-level spectral tokenization creates massive token count disparity ($H \times W$ vs. $H/P \times W/P$).

Solutions Implemented:

1. **Differentiated Compression Ratios:** Different keep ratios for spatial ($\sim 15\%$) vs. spectral ($\sim 0.8\%$) tokens
2. **Hierarchical Processing:** Multi-scale token division reduces computational load progressively
3. **Balanced Target Budgets:** Fixed final token counts regardless of initial imbalance
4. **Adaptive Scoring:** Modality-specific importance criteria account for different token characteristics

9.2 Cross-Modal Synchronization

Challenge: Different tokenization granularities could break spatial alignment between modalities.

Solutions Implemented:

1. **Shared 2D Positional Encoding:** Common spatial reference frame across tokenization scales
2. **Position-Aware Cross-Attention:** Spatial proximity bias in attention computation
3. **Joint Pair Selection:** Maintains bidirectional cross-modal relationships
4. **Spatial Coherence Loss:** Additional loss term encouraging spatially coherent selections

9.3 Computational Complexity

Challenge: Cross-attention between different token counts creates large attention matrices.

Solutions Implemented:

1. **Multi-Scale Processing:** Reduces effective token counts at coarser scales
2. **Progressive Compression:** Early token reduction minimizes downstream computational overhead
3. **Sparse Attention Patterns:** Optional windowed or local attention for efficiency
4. **Memory-Efficient Implementation:** Gradient checkpointing and mixed precision training

Complexity Analysis:

- **Original:** $O((HW)^2 + N_{\text{patches}}^2)$ for full cross-attention
- **After Compression:** $O(K_{\text{spectral}} \cdot K_{\text{spatial}})$ where $K_{\text{spectral}} \ll HW$
- **Compression Ratio:** Approximately 95% reduction in attention computation

10 Experimental Validation Framework

10.1 Dataset Specifications

The Token Cross-Modal Enhancer is designed for placenta HSI dataset with:

- **Image dimensions:** 512×512 pixels (varying by sample)
- **Spectral bands:** 38 bands in 515–700nm range (reduced to 30 effective bands)
- **Tissue classes:** Artery, Stroma, Vein, ICG regions, Blue/Red dyes
- **Annotation quality:** Medical doctor and student annotations with quality assessment

10.2 Evaluation Metrics

Segmentation Performance:

- Pixel-wise accuracy, mean IoU, class-specific F1 scores
- Boundary delineation quality (boundary IoU with different tolerance levels)
- Cross-modal fusion effectiveness through ablation studies

Computational Efficiency:

- Token compression ratios and accuracy preservation
- Memory usage and inference time comparisons
- FLOPs reduction analysis across different stages

Cross-Modal Analysis:

- Attention visualization across spatial-spectral token pairs
- Token selection quality assessment through importance score analysis
- Ablation studies on different compression strategies and scoring criteria

11 Implementation Details

11.1 Architecture Hyperparameters

- **Embedding dimension:** $D = 256$
- **Attention heads:** $H = 8$ per attention layer
- **Patch size:** $P = 16$ for spatial tokenization
- **Multi-scale factors:** $\{1, 2, 4\}$ with corresponding pooling
- **Temperature initialization:** $\tau_k = 1.0$ for all criteria

11.2 Training Configuration

- **Optimizer:** AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$
- **Learning rate:** Initial LR = 2×10^{-4} with cosine annealing
- **Batch size:** 16 for training, 32 for inference
- **Regularization:** Label smoothing ($\epsilon = 0.1$), dropout (0.1)
- **Loss function:** Cross-entropy + spatial coherence term

12 Conclusion

The Token Cross-Modal Enhancer presents a comprehensive solution for hyperspectral image segmentation that effectively balances computational efficiency with information preservation. The architecture’s key innovations include:

1. **Modality-Specific Tokenization:** Patch-based spatial and pixel-based spectral representations optimized for their respective characteristics
2. **Multi-Scale Cross-Modal Attention:** Comprehensive interaction patterns across different resolutions and modalities with hierarchical processing
3. **Intelligent Token Compression:** Multi-criteria scoring with cross-modal alignment preservation through two-stage optimization
4. **Robust Integration Pipeline:** Seamless connection with downstream processing while maintaining spatial correspondence

The system addresses fundamental challenges in hyperspectral processing through innovative token management strategies while maintaining the flexibility required for medical imaging applications demanding both spatial precision and spectral discrimination. The Token Cross-Modal Enhancer represents a significant advance in transformer-based architectures for dual-modal processing, offering both theoretical rigor and practical efficiency.

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